Keysight PROPSIM

Input Phase Calibration Based On External Signal

Application Note

KEYSIGHT TECHNOLOGIES

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1 INTRODUCTION

Phase calibration based on an external signal is an integrated solution of PROPSIM's test system for LTE/5G Over-the-air (OTA) scenarios. The phase calibration compensates the phase difference between the LTE streams outputted from the communication tester or base station emulator (BSE). The streams can have a random phase offset depending on e.g. the BSE unit, LTE band and cell state.

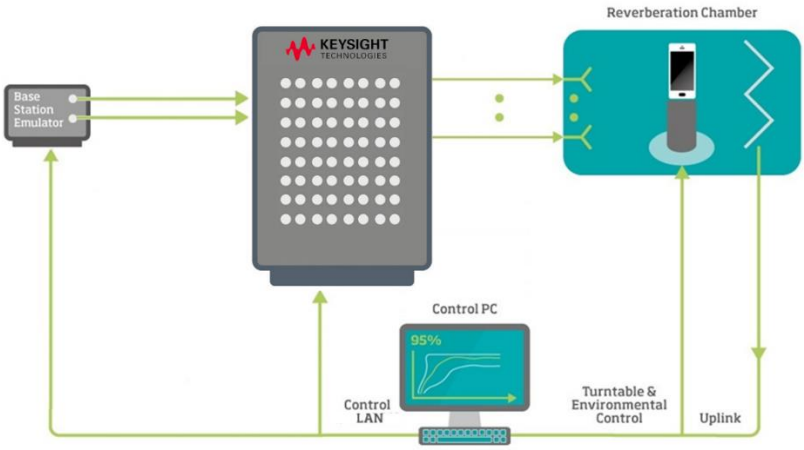

Figure 1 Example of RC (Reverberation Chamber) based set up

During the phase calibration process PROPSIM detects the presence and validity of the input LTE/5G signals. If valid LTE/5G signal is detected the phase difference between the input signals is analysed. Finally, the PROPSIM input phases are adjusted to compensate the measured phase differences.

Phase calibration based on an external signal can be used in OTA test set up. Both anechoic and reverberation chamber based methods are supported. Example connection of reverberation chamber based set up is shown in Figure 1.

1.1 HW and SW requirements and specifications

Table 1 HW and SW requirements

Parameter	Value
PROPSIM HW	PROPSIM F64, PROPSIM FS16
PROPSIM firmware	Version 1.0 or above Version 9.0 or above (5G)
Number of supported inputs	2 (LTE) 2..64 (Correlated signal) 2, 4 or 8 (5G)
Supported technologies	LTE, Correlated signal, 5G
Supported LTE parameters	Cell ID: 0,2,...,504 Bandwidth: 5,10,20 MHz
Supported 5G parameters	Bandwidth: 20, 40, 50, 100 MHz Cell ID: 0..1007 Sub carrier spacing: 15 kHz, 30 kHz, Automatic Requirements: Cell load: Empty cell and no traffic Number of SSB beams: 1 SSB periodicity: 20 ms

Parameter	Value
Input phase accuracy after calibration	± 5º (typical)

1.2 Document History

The following table lists the main changes to issues of this document.

Issue	Date	Summary of Changes
1.0	Jan 2015	Release of PROPSIM – Application Note Input Phase Calibration based on external signal
1.1	Jan 2016	Template updated, PROPSIM release 3.7 updates
1.2	Sep 2016	Package version 1.2 - Release notes chapter added
1.3	Apr 2017	Package version 1.3 – See release notes chapter for details
1.4	Jan 2018	Rebranded using Keysight template.
1.5	Jun 2019	PROPSIM F64 changes
1.6	Jun 2021	5G phase tuning added
1.7	Jan 2024	5G phase tuning improvements

2 ATE INTERFACE

2.1 ATE commands

2.1.1 SYSTem:CALIBration:INPut:PHASETune?

Calibrates input phases based on external signal

Syntax:

SYSTem:CALIBration:INPut:PHASETune? <Number of inputs (N)>,<input id#1>,...,<input id#N>,<Automatically adjust phases>,<External signal technology>,<Technology Specific parameter #1>,...,<Technology Specific parameter #M>

Description:

This command calibrates input phases based on external signal. Measured phases are reported out as a reply. If auto adjust is chosen, values are automatically set to phase settings of PROPSIM inputs.

Automatically adjust phases:

0, no auto adjustment
1, measured phase differences are automatically set to phase settings of inputs

External signal technology:

LTE, External signal technology LTE
5G, External signal technology 5G
None, External signal technology not specified. Uses signal coherence.

LTE technology specific parameters:

Parameter#1: External Signal bandwidth in MHz (5, 10 and 20MHz are supported)
Parameter#2: Cell Id (integer number)
Parameter#3: Signal center frequency in MHz

5G technology specific parameters:

Parameter#1: External Signal bandwidth in MHz (20, 40, 50 and 100 are supported)
Parameter#2: Cell Id (integer number)
Parameter#3: Signal center frequency in MHz
Parameter#4: (optional): Sub carrier spacing (SCS) in kHz (15 and 30 are supported, 0=auto detect)

None technology specific parameters:

Parameter#1: External Signal bandwidth in MHz
Parameter#2: Signal center frequency in MHz

Calculation successful response:

1,<Phase calibration value for input id#1 in degrees>, ... ,<Phase calibration value for input id#N in degrees>

Calculation unsuccessful response:

0,<Reason for failing>

Note: *First given input is used as an reference input (phase calibration value is always 0)*

Example 1 (LTE):

// start input phase calibration with auto adjust. External signal technology is LTE, Bandwidth is 10MHz, signal center frequency is 2200 MHz, cell Id is 1, tuned inputs are 1 and 2. Calibration values are 0 degrees to input 1 and -10 degrees to input 2.

syst:calib:inp:phaset? 2,1,2,1,LTE,10.0,1,2200

1,0,-10

Example 2 (5G):

// start input phase calibration with auto adjust. External signal technology is 5G, Bandwidth is 100MHz, signal center frequency is 2050 MHz, cell Id is 6, tuned inputs are 1, 2, 3 and 4. Calibration values are 0 degrees to input 1,-20 degrees to input 2, 30 degrees to input 3, -40 degrees to input 4.

syst:calib:inp:phaset? 4,1,2,3,4,1,5G,100.0,6,2050

1,0,-20,30,-40

Example 3 (none, coherent signal in all inputs):

// start input phase calibration with auto adjust. External signal technology is None, Bandwidth is 40MHz, signal center frequency is 5000MHz, tuned inputs are 1, 2, 3 and 4. Calibration values are 0 degrees to input 1,-20 degrees to input 2, 30 degrees to input 3, -40 degrees to input 4.

syst:calib:inp:phaset? 2,1,2,3,4,1,None,40.0,5000

1,0,-20,30,-40

2.2 ATE Errors

Table 2 ATE errors

Error	Explanation
Given technology not supported	Given technology is not supported or technology is misspelled.
Given bandwidth not supported	Given Bandwidth is not supported.
Channel input not found	Given input is not part of emulation or same input is given twice.

2.3 Failing reasons

Table 3 Phase adjustment failure reasons and explanations

Fail	Explanation
Given Bandwidth not supported	Check bandwidth.
Given signal count not supported	Given input count not supported.
Defined signal out of emulator bandwidth	Check given frequency and bandwidth.
Some of the needed parameter missing	Check given parameters.
Given Cell Id not supported	Check given cell id.
Differences in center frequencies of the inputs	Given inputs have different center frequencies.
Measurement package version not supported / Too old Signal Analysis Package installed	Firmware and PROPSIM Signal Analysis Package version conflict. Contact Keysight.
Measurement package missing / Signal Analysis Package not installer	PROPSIM Signal Analysis Package missing. Contact Keysight.
Signal not recognized	Check signal and retry.

Fail	Explanation
Signal missing	Check signal level and retry.
Pilot signal not found	Check signal, given parameters and retry (LTE).
SSB signal not found	Check signal, given parameters and retry (5G).
Record failed	Internal error. If error consists contact Keysight.
SW Restriction	Current version of PROPSIM Signal Analysis Package or PROPSIM firmware version does not support given measurement setup.